

DrugLens

Geriatric Polypharmacy Safety Auditor

A deterministic clinical rules engine that runs on every request in ~3 ms — escalating to a cloud model only when the case actually warrants it.

102

curated DDIs

61

Beers 2023

58

STOPP/START v3

~3 ms

engine latency

0

tokens on safe cases

75

tests passing

Live: <http://172.105.40.61:8000> Repo: github.com/Vishwazeer/DrugLens

The Geriatric Polypharmacy Crisis

Polypharmacy — five or more daily prescriptions — is now the norm in elderly care. It is also dangerous.

- **40%+ of seniors** take five or more daily prescription medications.
- **~30% of emergency admissions** in adults 65+ are driven by adverse drug events.
- **\$3.5 billion** in avoidable US emergency-care spend every year.
- **Renal and hepatic decline** reduce clearance — standard adult dosing becomes toxic in the elderly.
- **Falls, delirium, kidney injury** are the downstream cost: benzodiazepines, anticholinergics, the NSAID 'triple whammy'.

The gap:

Standard clinical workflows do not audit age-specific risk. The tooling that exists makes it worse.

Why Current EHR Systems Fail

The problem is not that checkers miss interactions. It is that they flag everything.

I. Alert Fatigue

Checkers flag every theoretical drug pair regardless of severity or context. Clinicians mute or ignore up to 90% of warnings — so the life-threatening ones get lost in the noise.

II. No Age-Specific Rules

Standard DDI checkers ask 'does A interact with B?' They never ask whether drug A is inappropriate for an 85-year-old at all, or whether the patient's eGFR forbids it.

III. Unstructured Clinical Notes

Much of the medication record lives as free-text prose, not clean tables. Re-typing it into a checker costs clinician time nobody has.

IV. The 'Novel Drug' Blindspot

Lookup databases only know pre-indexed pairs. A new or uncommon combination simply returns nothing — indistinguishable from 'safe'.

The Solution: Precision, Not Volume

Deterministic rules always run first. The LLM is invoked only when the case earns it.

Free-text note OR medication list

Clinical input

v

Clinical Text Parser

drug-vocabulary anchored · lifts age / eGFR / conditions

v

Deterministic Rules Engine — ALWAYS RUNS · ~3 ms · 0 tokens

102 DDIs (combination-aware) · 61 Beers 2023 (age + eGFR gated) · 38 STOPP + 20 START

v

Token-Efficient Router

LOW / MINIMAL answered here — zero LLM tokens. Only MODERATE / HIGH escalate.

v

Cloud Model (Fireworks AI)

streaming narrative · prescribing alternatives · novel-DDI via PubChem SMILES

What Makes This Different

Three things a lookup table cannot do — each demonstrable live, each locked by tests.

1 True combination logic (AND-semantics)

'Opioid + benzodiazepine' and the renal 'triple whammy' fire ONLY when every component is present — never on a single drug. This is the root fix for alert fatigue, and naive engines get it wrong.

PROOF: our safe demo case produces 0 alerts. Opioid alone: silent. Opioid + benzo: fires.

2 Age- and eGFR-gated clinical rules

Beers and STOPP rules are gated on the patient's real age, comorbidities and renal function — not applied blindly.

PROOF: same drugs, age 30 -> 0 Beers. Age 85 -> 2. Metformin @ eGFR 90 -> silent. @ eGFR 25 -> flagged.

3 Token-efficient routing

Low-risk patients are fully answered by the offline engine. The cloud model is never called.

PROOF: 2 of our 3 demo cases spend ZERO LLM tokens end to end.

Closing the 'Novel Drug' Blindspot

A lookup table returns nothing for an un-indexed pair — indistinguishable from 'safe'. We don't stop there.

Every drug pair with NO database entry is escalated — grounded in real chemistry.

1. Identify all pairs the curated database has no entry for.
2. Fetch each drug's molecular SMILES structure from PubChem.
3. One batched, structure-grounded call to the cloud model — JSON-mode, severity-scored.
4. Findings scored 'none' are discarded, so we never recreate the alert fatigue we set out to fix.

Live output — interactions no lookup table in this project contains:

sertraline + mirtazapine

Serotonin syndrome risk — additive serotonergic activity

MODERATE

SMILES-grounded

sertraline + tamsulosin

CYP2D6 inhibition raises tamsulosin exposure — hypotension, fall risk

MODERATE

SMILES-grounded

Honest Scope

Being precise about this is worth more to us than a bigger-sounding claim.

Deterministic clinical engine (102 DDI / 61 Beers / 58 STOPP-START)

LIVE

LIVE — runs on every request, before and independently of any LLM. ~3 ms, measured.

Token-efficient routing (0 LLM tokens on low-risk cases)

LIVE

LIVE — 2 of 3 demo cases never invoke a GPU at all.

Cloud synthesis — narrative, alternatives, novel-DDI prediction

LIVE

LIVE — via Fireworks AI (deepseek-v4-pro, configurable).

MedGemma 4B / TxGemma 2B on AMD Instinct (vLLM + ROCm)

NOT RUN

SHIPPED, NOT ENABLED — our team's Instinct pod stayed stuck in 'Creating...' all window. The code path is real and reviewable (setup_amd_pod.sh, gpu Compose profile) but is unbenchmarked, and we make NO performance claim for it.

Every number the app displays is read live from the loaded rulesets (GET /api/engine-stats) — never hardcoded. Anything we claim can be checked against the repository.

Why DrugLens

What we would like to be judged on.

Attacks the real problem

Alert fatigue is a precision problem, not a coverage problem. Combination AND-semantics is the fix — and almost nobody implements it.

Clinically grounded

AGS Beers 2023 and STOPP/START v3 — the actual reference standards, not an invented ruleset.

The rules cannot be hallucinated

The deterministic engine always runs first. The LLM only narrates findings it can neither invent nor suppress.

Verifiable, not asserted

75 offline tests green in CI. Every displayed statistic read live from the rulesets. Honest scope, stated up front.

Complete and deployed

React 19 + FastAPI, single-origin, containerised, live and open — with PDF clinical export.

Live: <http://172.105.40.61:8000>

Repo: github.com/Vishwazeer/DrugLens

Decision-support prototype. Not a medical device; not clinically validated; not for real prescribing decisions.